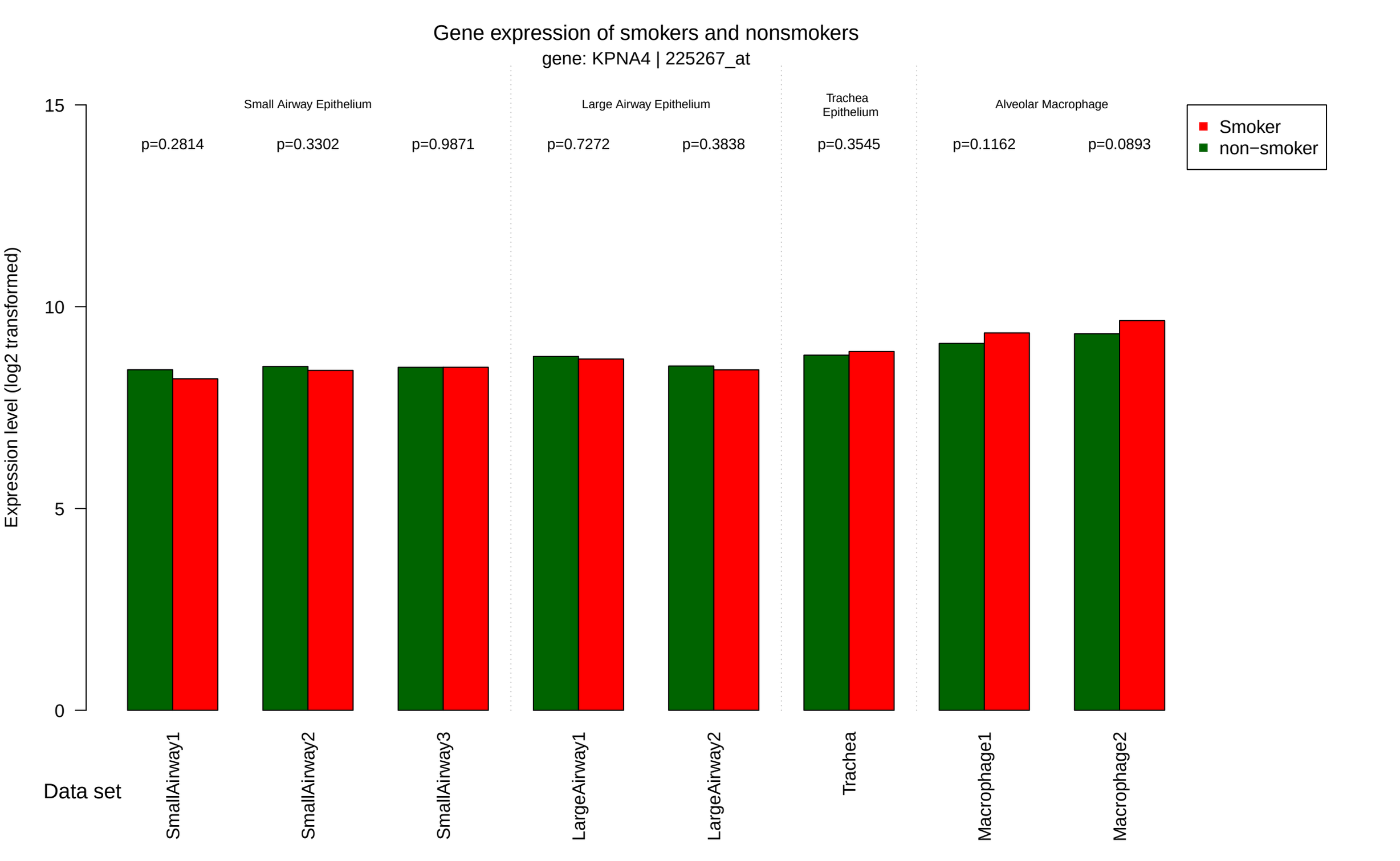
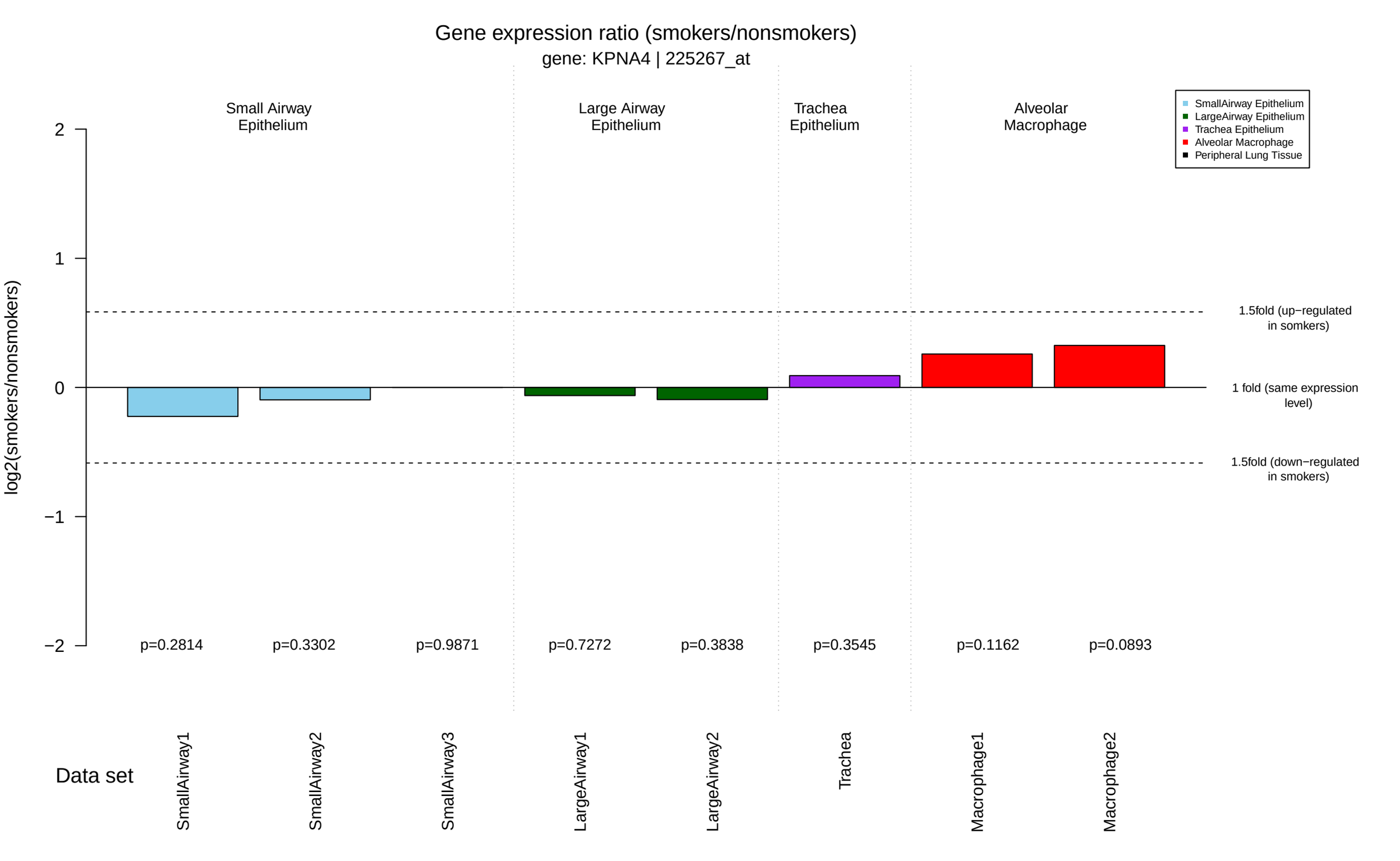
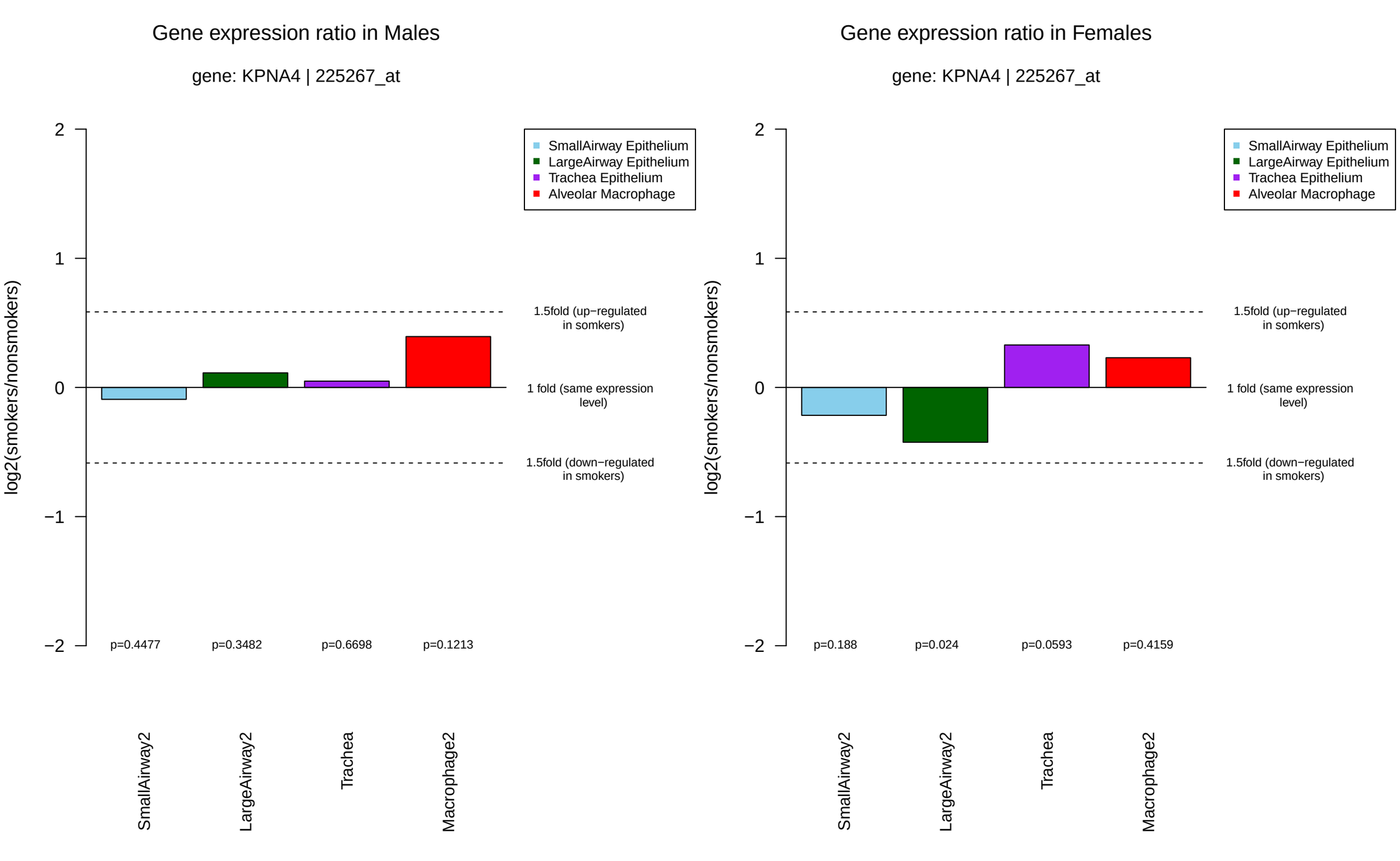
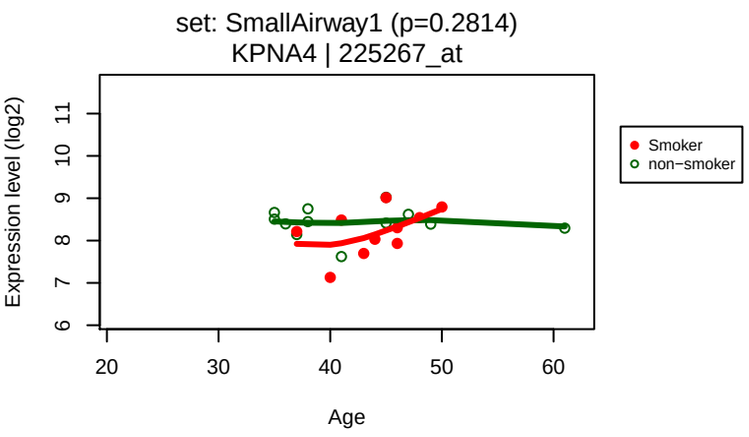
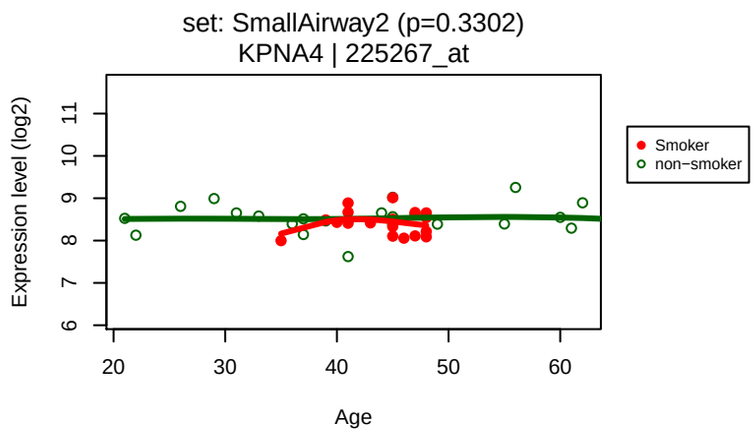
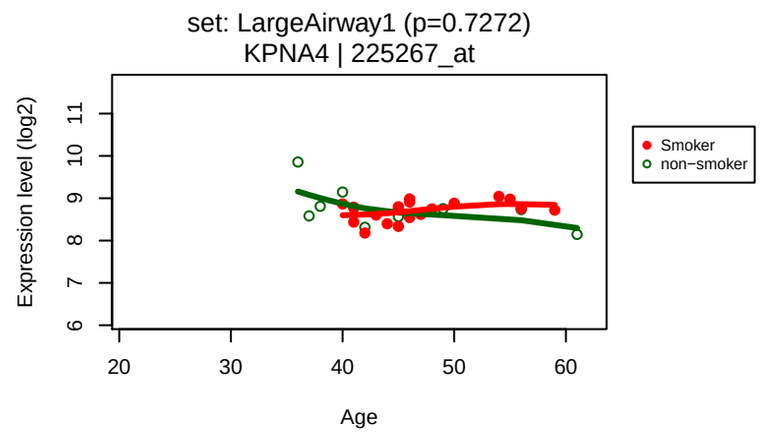

Gene expression of smokers and nonsmokers
gene: KPNA4 | 225267_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2814
p=0.3302
p=0.9871
p=0.7272
p=0.3838
p=0.3545
p=0.1162
p=0.0893
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

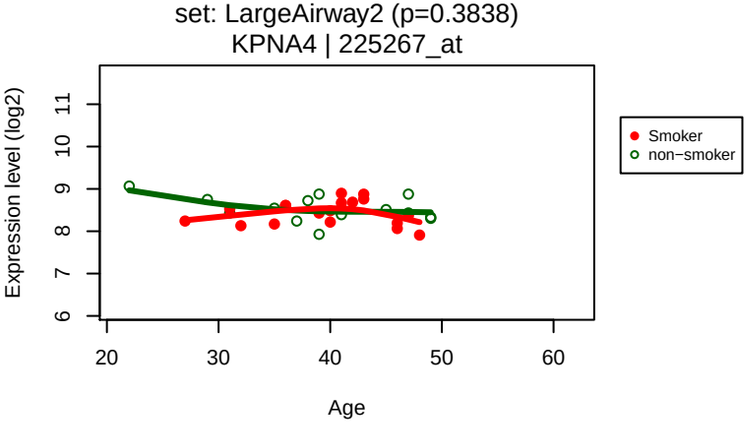

Gene expression ratio (smokers/nonsmokers)
gene: KPNA4 | 225267_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2814
p=0.3302
p=0.9871
p=0.7272
p=0.3838
p=0.3545
p=0.1162
p=0.0893
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

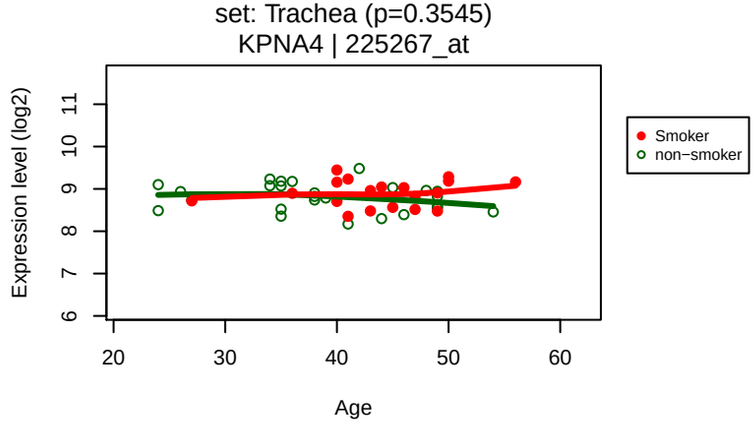

Gene expression ratio in Males
gene: KPNA4 | 225267_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4477
p=0.3482
p=0.6698
p=0.1213
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KPNA4 | 225267_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.188
p=0.024
p=0.0593
p=0.4159
SmallAirway2
LargeAirway2
Trachea
Macrophage2

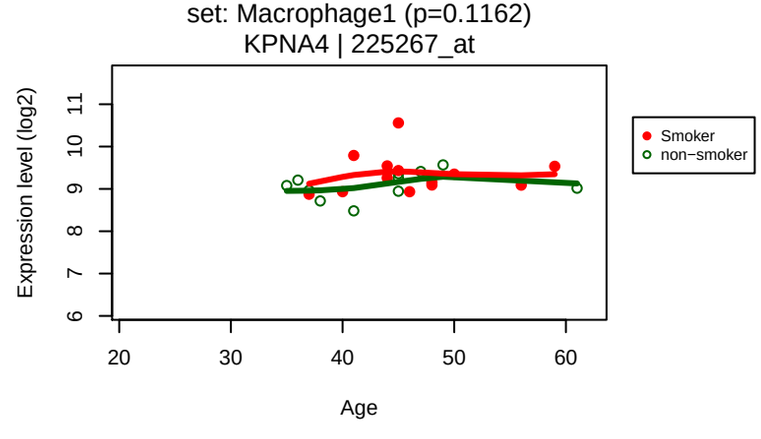

set: SmallAirway1 (p=0.2814)
KPNA4 | 225267_at
Expression level (log2)
Age
Smoker
non−smoker

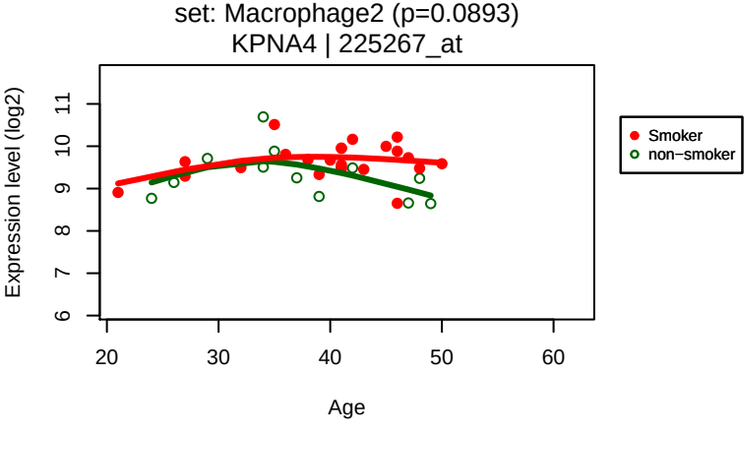

set: SmallAirway2 (p=0.3302)
KPNA4 | 225267_at
Expression level (log2)
Age
Smoker
non−smoker

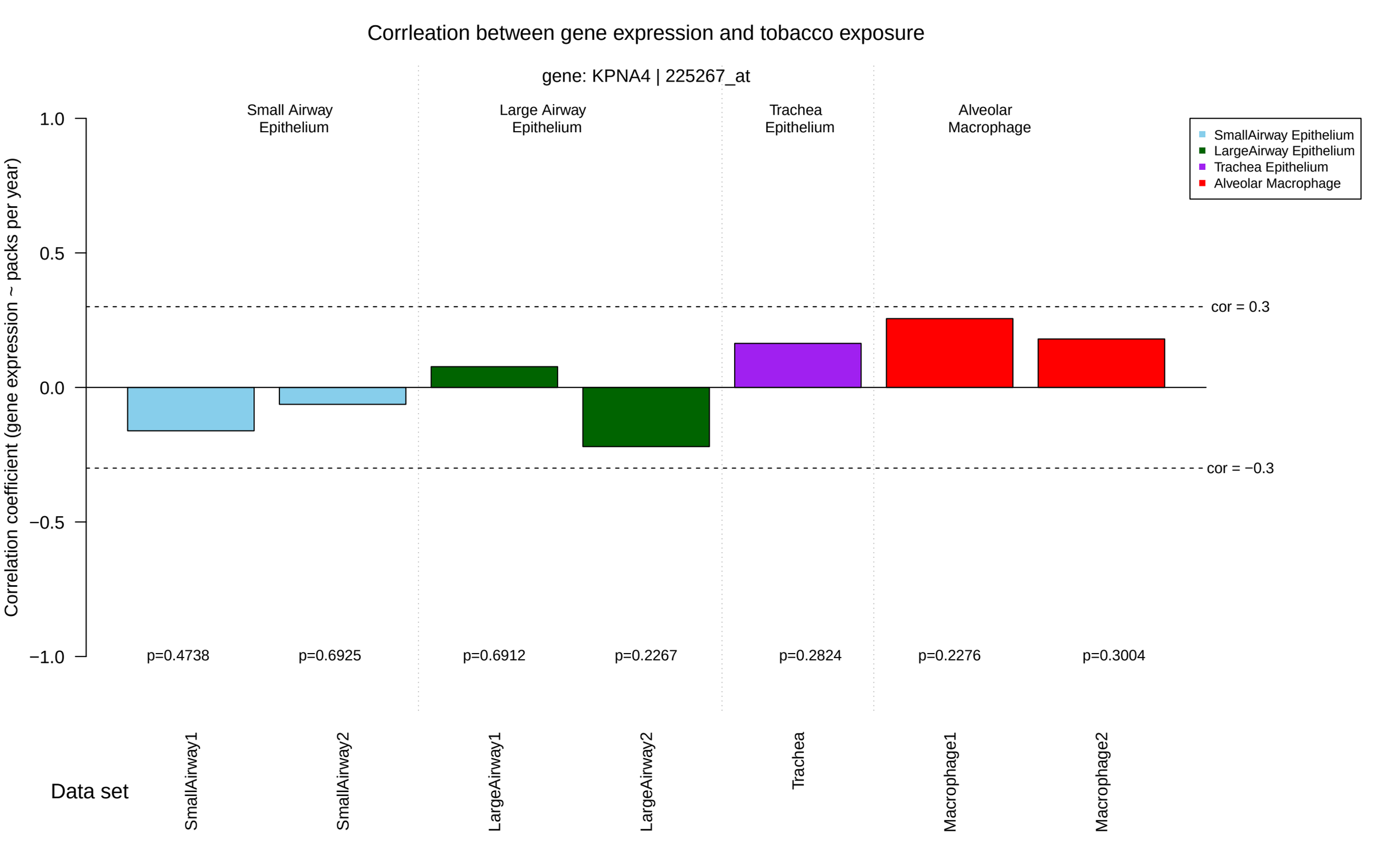

set: LargeAirway1 (p=0.7272)
KPNA4 | 225267_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3838)
KPNA4 | 225267_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3545)
KPNA4 | 225267_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1162)
KPNA4 | 225267_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0893)
KPNA4 | 225267_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KPNA4 | 225267_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4738
p=0.6925
p=0.6912
p=0.2267
p=0.2824
p=0.2276
p=0.3004
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2